



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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FIRST REVISION EXAMINATION - JANUARY - 2025

BIOLOGY

Time Allowed : 3.00 Hours]

Part - I (Bio-Botany) (Marks : 35)

[Max. Marks : 70

Section - I

Note. (i) Answer All the questions.

8×1=8

(ii) Choose the most appropriate answer from the given four alternatives and write the option code and corresponding answer.

- Which mendelian idea is depicted by a cross in which F₁ generation resembles both the parents
 - Incomplete dominance
 - Law of dominance
 - Inheritance of one gene
 - Co-dominance
- Select the incorrect statement from given statement
 - A tonic used for cardiac arrest is obtained from Digitalis Purpurea
 - Medicine used to treat Rheumatic pain is extracted from Capsicum annum
 - An anti malarial drug is isolated from cinchona officinalis
 - Anti-Carcinogenic property is not seen in Catharanthus roseus
- Match the following

i) External fertilization	-	(i) Pollen grain
ii) Androecium	-	(ii) Anther wall
iii) Male gametophyte	-	(iii) Algae
iv) Primary parietal layer-	(iv) stamens	

 - I - iv, II - i, III - ii, IV - iii
 - I - iii, II - iv, III - i, IV - ii
 - I - iii, II - iv, III - ii, IV - i
 - I - iii, II - i, III - iv, IV - ii
- Assertion (A)** : Parthenocarpy which arises due to mutation is called genetic parthenocarpy
Reason (R) : Seedless fruits are useful for the preparation of jams.
 - Both assertion & Reason are true & Reason is correct explanation of Assertion
 - Both Assertion & Reason are true but reason is not correct explanation of Assertion
 - Assertion is true; Reason is false
 - Both Assertion & Reason are false
- Select the incorrect Pair
 - Trisomy - Datura
 - Triploid - Banana
 - Triticalc - High Valine
 - Translocation - Non-homologous chromosome
- An antibiotic gene in a vector usually helps in the selection of
 - competent cells
 - Transformed cells
 - Recombinant cells
 - None of the above
- Virus free plants are developed from
 - Organ culture
 - Meristem Culture
 - Protoplast culture
 - Cell suspension culture
- The number of linkage groups in maize is
 - 15
 - 20
 - 25
 - 10

SECTION - II

II. Answer any four questions.

4×2=8

9. Write short note on Pollen Kitt.

10. Distinguish tenuinucellate & Crassinucellate Ovules.

11. Define true breeding.

13. What is meant by Bio-Piracy?

12. Explain the Phenomenon of synapsis.

14. Explain Organogenesis.

SECTION - III

III. Answer any three questions. Question No. 19 is compulsory.

3×3=9

15. How will you identify a Vector?

16. Write note on types of linkage.

17. What do you know about Germplasm conservation. Describe it.

18. Explain with an example how single gene affect multiple traits & alleles the phenotype of an Organism.

19. Draw a neat labelled diagram of Polyembryony Embryo Sac.

IV. Answer all the following Questions:

2×5=10

20. (a) What is bioremediation? Give some examples of bioremediation.

(OR)

(b) Describe Dominant Epistasis with an example.

21. (a) Enumerate the Characteristic features of Entomophilous flowers.

(OR)

(b) Explain the basic concepts involved in Plant Tissue Culture.

CH / J / 12 / Bio / 1

SATGURU STUDY CENTRE

PART - II (BIO-ZOOLOGY) (Max. Marks: 35) SECTION - I

Note: (i) Answer all the questions

(ii) Choose the most appropriate answer from the given four Alternatives and write the option code and the corresponding answer

8x1=8

1. The Purpose of Biological Treatment of waste Water is to -----
a) Reduce BOD
b) Increase BOD
c) Reduce Sedimentation
d) Increase Sedimentation
2. A Population will not exist in Hardy Weinberg equilibrium if
a) Individuals mate Selectively
b) There are no Mutations
c) There is no migration
d) The Population is large
3. Which of the following is not Correct?
a) Three or more alleles of a trait in the population are called multiple alleles.
b) A normal gene undergoes mutations to form many alleles
c) Multiple alleles map at different loci of a chromosome
d) A diploid organisms has only two alleles out of many in the population
4. Continuous breeders continue to breed Throughout their Sexual maturity eg
a) Lizards
b) Frogs
c) Deers
d) Honey bees
5. Find the Wrongly Matched pair
a) Bleeding Phase - Fall in Oestrogen and Progesterone
b) Follicular phase - rise in Oestrogen
c) Luteal Phase - Rise in FSH Level
d) Ovulatory phase - LH surge
6. Identify the correct Statements from the following
a) Chlamydiasis is a viral disease.
b) Gonorrhoea is caused by a spirochete bacterium, Treponema palladium.
c) The incubation period for syphilis is 2 to 14 days in males and 7 to 21 days in females.
d) Both syphilis and gonorrhea are easily cured with antibiotics.
7. Meselson and stahl's experiment proved
a) Transduction
b) Transformation
c) DNA is the genetic Material
d) Semi - Conservative Nature of DNA replication
8. Match the Pathogens with respective disease caused by them and Select the Correct match using the Codes given below
a) Leishmania Donavani - i) Amoebiasis
b) Wuchereria Bancrofti - ii) Kala - Azas
c) Trypanosoma Gambiense - iii) Sleeping Sickness
d) Entamoeba Histolytica - iv) Filariasis
a) A - ii, B - iv, C - iii, D - i
b) A - ii, B - iv, C - iii, D - i
c) A - iii, B - i, C - ii, D - iv
d) A - i, B - iv, C - iii, D - ii

SECTION - II

II. Answer any four questions. (Q. No: 14 is Compulsory)

4x2=8

9. Expand the Following: a) ZIFT b) ICSI
10. Why is the Offspring formed by Asexual reproduction referred as a clone?
11. What are Holandric genes?
12. What is Bioremediation?
13. What is Vaccine? what are its types?
14. List out the Major gases Seems to be found in the Primitive Earth.

SECTION - III

III. Answer any 3 questions. (Question No. 19 is compulsory)

3x3=9

15. What are the Interferons? Mention their Role?
16. Differentiate between the Autogamy and Exogamy?
17. Placenta is an Endocrine tissue. Justify.
18. What are the Application of Karyotyping?
19. State any three goals of the Human genome Project.

SECTION - IV

IV. Answer in Detail.

2x5=10

20. a) Give a Schematic Representation of Spermatogenesis and Oogenesis in Human. (OR)
b) Give an account of Genetic Control of RH Factor.
21. a) Explain how mutations, natural Selection and Genetic drift affect Hardy weinberg equilibrium.
(OR)
b) Differences between Active and Passive Immunity.